

Image Classification from Randomized and Deterministic EEG Trials

Anonymous CVPR submission

Abstract

001 In this paper, we propose a new deep model, TS-
002 SpectrumNet, to achieve improved image classification from
003 both randomized and deterministic EEG trials, which is ex-
004 tremely challenging and the best classification rate reported
005 so far is only 7.0% for randomized trials and 48.1% for de-
006 terministic trials. To confront the challenge brought in by
007 randomized EEG trials and semantics ambiguity inside nat-
008 ural stimuli images, we propose two new modules: TSDL
009 (Temporal-Spatial Dynamics Learning) and SpecRL (Spec-
010 trum Representational Learning) to achieve the best pos-
011 sible exploitation of the classifiable information embedded
012 inside EEG trials. While the first module is introduced to
013 simultaneously learn temporal-spatial dynamics with multi-
014 ple convolutional kernels to accommodate a broader range
015 of varying characteristics of both the randomized and de-
016 terministic EEG trials, the second module is introduced to
017 achieve representational learning in spectrum domain to
018 characterize those hierarchical global correlations beyond
019 the constraint of localities. In this way, our proposed TS-
020 SpectrumNet is able to achieve not only the comprehen-
021 sive capture of complex dynamics inside EEG sequences
022 but also the full exploitation of spectrum features across all
023 temporal-spatial EEG channels. To minimize the impact of
024 noises, we further introduce a channel-wise normalization
025 scheme to optimize the quality of EEG input for our pro-
026 posed TS-SpectrumNet. Extensive experiments are carried
027 out and the results validate that: (i) over randomized EEG
028 trials, our proposed delivers a classification rate of 15.1%,
029 more than doubling the best rate reported so far; and (ii)
030 over deterministic EEG trials, our proposed delivers the
031 best classification rate of 50.3% at the cost of 0.1 million
032 parameters, whilst the existing best performing EEGChan-
033 nelNet achieves the classification rate at 48.1% at the cost
034 of 5 million parameters.

1. Introduction

036 At present, EEGs have been widely adopted for analyzing
037 brain states in psychology, and their research is primarily
038 dominated by stimulations of spiky patterns or object im-

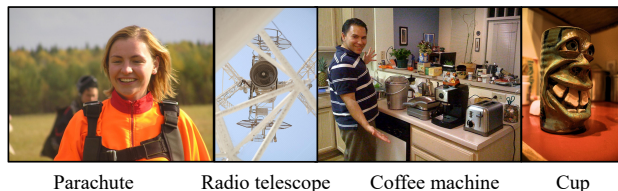


Figure 1. Illustration of stimuli image samples used for EEG-ImageNet [1, 25].

039 ages [2, 8, 11, 13, 29]. In 2017, Spampinato et al. [25]
040 selected 40 classes of natural images from ImageNet [21]
041 to stimulate human brains and then classify brain responded
042 EEG trials to recognize image categories, leading to the cre-
043 ation of EEG-ImageNet, the world’s first EEG dataset gen-
044 erated in an environment close to realistic mind-reading sce-
045 narios. As this data set is created by stimulations of natural
046 images in a deterministic order, we name it ‘EEG-ImageNet
047 with deterministic trials’ to distinguish from the latter work
048 on randomized EEG trials [1]. With such deterministic
049 EEG-ImageNet, an encouraging classification rate of 82.9%
050 is reported [25], which ignited significant interests across
051 relevant research communities [3, 5, 9, 12, 19, 20, 32, 33].
052 When EEG-ImageNet is recreated by Ahmed et al. [1] with:
053 (i) each of the 40 classes is represented by 10 times more
054 images, totaling 40,000 natural images; and (ii) the stimula-
055 tion of all natural images is arranged in randomized orders,
056 however, the classification rate is significantly dropped and
057 the maximum rate achieved is reported to be 7.0% [1, 17].
058 To the best of our knowledge, no work has been reported so
059 far to move any further beyond this rate.

060 Compared with stimuli of spiky patterns or object im-
061 ages, stimuli with natural images are often content-rich
062 with multiple semantics, making it more difficult for human
063 brains to achieve desirable responses due to the ambiguities
064 created by such multi-semantics and their mixed presence.
065 Figure 1 illustrates four samples together with their class la-
066 bels, selected from the stimuli images for EEG-ImageNet.
067 While the first sample is labeled as the class ‘Parachute’,
068 for example, it is difficult for human eyes to see the con-
069 nection with ‘Parachute’ and recognize this class from the

sample image only, let alone that the brain response is collected within its stimulation period (2 seconds for EEG-ImageNet). Other sample images present similar problems as seen from Figure 1.

To advance this direction of research, i.e classification of those natural stimuli images directly from both randomized and deterministic EEG trials, we propose a dedicated deep framework, TS-SpectrumNet, to address all the challenges described above. In comparison with the existing state of the arts, our proposed TS-SpectrumNet features in a comprehensive learning and tracking of EEG trials in terms of temporal-spatial dynamics, spectrum representations, and depth revelations. We highlight our contributions as follows:

- To combat the additional noises introduced by both the randomized EEG trials and the class labeling ambiguities embedded inside the stimuli with natural images, we propose a new scheme of channel-wise normalization to improve the quality of raw EEGs, and thus making it more presentable to our proposed TS-SpectrumNet;
- To increase the representational capacity and comprehensiveness of our proposed TS-SpectrumNet, we propose to add a temporal-spatial dynamics learning module and a spectrum representational learning module to maximize the exposure of classifiable information for both the randomized and deterministic EEG trials;
- While the temporal-spatial dynamics learning module is tailored to maximize the temporal/spatial exposure of the classifiable information through multi-scale and multi-kernel convolutions and thus optimizing their capture of complex dynamics embedded inside the EEG sequences, the proposed spectrum representational learning features in multi-level Wavelet decomposition of the input to enable the proposed deep model to: (i) extract hierarchical spectrum features to sustain global characteristics and overcome local constraints; (ii) optimize its comprehensiveness in exposing and capturing the classifiable information out of EEG sequences, and hence significantly increase the deep model’s capability in classifying natural stimuli images from EEG trials;
- Extensive experiments and ablation studies are carried out, the results of which not only validate the effectiveness of our proposed TS-SpectrumNet, but also illustrate that TS-SpectrumNet outperforms all the existing representative baselines, and for classification of randomized EEG trials, TS-SpectrumNet pushes the existing maximum rate, 7%, to a new level of 15.1%, creating an encouraging potential for further advances along this line of research. For classifying brain perception of natural images from deterministic EEG trials, our proposed achieves the best rate of 50.3% at the cost of 0.1 million parameters, whilst the best performing EEGChannelNet delivers 48.1% with 5 million parameters.

2. Related works

Introduced by Li and Ahmed et al. [1, 17], classification of randomized EEG trials upon their extended EEG-ImageNet is tested by a range of classifiers, including not only those representative deep models, such as EEGNet [15], EEGChannelNet [20], LSTM [7], 1DCNN, and SyncNet [18] etc., but also those traditionally well-known classifiers, such as SVM [10], MLP [27], and KNN [30] etc. While the best rate of 7.0% is achieved by EEGNet and the second best rate of 5.1% is achieved by CNN, traditional classifier SVM surprisingly produced a classification rate of 5%, the third best, outperforming all other deep models, such as LSTM, SynNet, and EEGChannelNet.

EEGNet [15] is a portable and simple network with separable convolution stages to learn temporal summary of individual feature maps, followed by pointwise convolution, which learns how to optimally mix the feature maps together. Trained by a cross-entropy loss function, EEGNet relies on a softmax-based classifier to deliver the final classification results. EEGChannelNet, on the other hand, utilizes 1D convolution and incorporates residual blocks to enhance its classification performances. On classifying deterministic EEG trials, EEGChannelNet achieved the best classification rate of 48.1% [20]. When applied to classifying randomized EEG trials, however, its classification rate is dropped to 2.5% [1].

Given the fact that Transformers are notably strong for capturing long-term dependencies, recent research can also be characterized as integrating Transformers to tackle both temporal and spatial feature extraction via representational learning [16, 23, 26]. EEGConformer [23] remains to be one of such representative examples, which combines the strengths of CNNs and Transformers by integrating convolutional layers for local feature extraction with Transformer layers for capturing global dependencies, leading to improved performances over EEG classification tasks. Siddhad et al. [22], however, pointed out that the positional encoding in Transformers is not tailored for EEG data, which could lead to performance degradation in certain cases.

Another direction for classifying brain perception of images from EEGs is represented by the work pioneered by C. Du et al. [4] and Song Y. et al.[23, 24], which is based on the EEG data set collected by Gifford et al [6]. Specifically, this data set is collected from 10 subjects, to whom both the training images and test images are presented in a pseudo-randomized order. In total, the training set contains 1654 concepts, each of which includes 10 object images, totaling 16540 images, and the test set contains 200 concepts, each of which is represented by 1 object image. All images are presented to the subject for 100ms, followed by 100ms blanking, and EEGs are collected with 63 channels at a 1000Hz sampling rate. While the collection of EEGs are replicated 4 times over the training set, collection

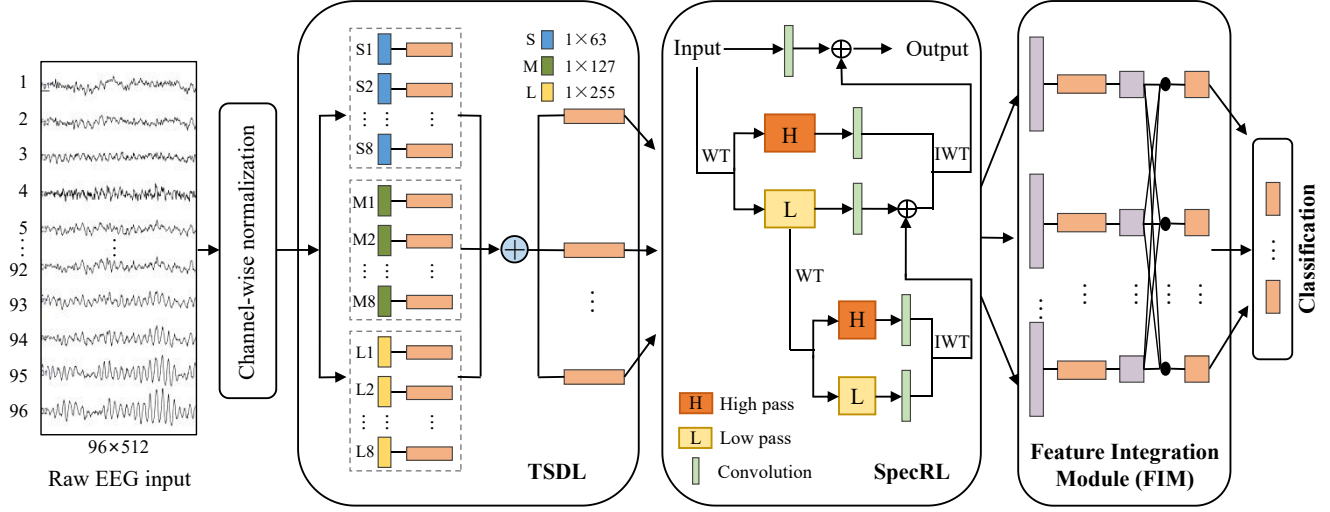


Figure 2. Structural overview of our proposed TS-SpectrumNet.(L: large convolution kernel; M: medium convolution kernel; S: small convolution kernel)

of EEGs over the test set is replicated 8 times to provide a richer EEG dataset for research purposes [6]. As this data set is essentially established for modeling human visual object recognition, it is impossible to classify the 1654 concepts directly from the collected EEGs, due to the fact that: (i) lack of training EEGs since each concept only contains 10 images to stimulate brains; (ii) there are too many concepts, 1654 in total, making it difficult to distinguish even directly from those stimulating images, let alone from those brain responded EEGs. To this end, Song et. al. introduced the concept of 'brain decoding' to describe the cognitive process of classifying brain perceptions [24], where a self-supervised framework, containing an EEG encoder, a stimulating image encoder, and a contrast learning module, is proposed. While image encoder and EEG encoder extract features from stimulating images and EEGs respectively, the contrast learning aligns these features to optimize their pairings. At the testing stage, object images that have not appeared in EEG collections (zero-shot) are averaged to formulate a single concept template, which are then processed together with test EEGs by the framework to calculate their match and produce a top-5 classification rate, if the correct match appears within any of the top 5 results, and top-1 classification rate, where the correct match only appears at the top. Comparative experiments are reported in [24] that, in terms of top-1 classification rates, their proposed decoding framework achieved 15.60% on average across all subjects, whilst the existing baseline by Du et al. [4] only achieved 5.8% on average across all subjects, providing an additional indication that classification of randomized EEG trials is indeed extremely challenging even under the contexts that corresponding stimulating images are exploited to provide essential match and pairing.

3. Our proposed TS-SpectrumNet

Figure 2 illustrates a structural overview of our proposed TS-SpectrumNet, which can be seen to include four operational stages, including: (i) channel-wise normalization stage to enhance the quality of the input raw EEG sequences; (ii) temporal-spatial dynamics learning module to provide more comprehensive coverage and tracking for the complex time-dependent dynamics across EEG sequences; (iii) spectrum representational learning module to sustain Wavelets-based multi-resolution and multi-level decomposition of the information flow across channels and thus optimize the exploitation of global correlations beyond local constraints and the coordination/interaction among all channels; and (iv) a feature integration module (FIM) that adaptively fuses the outputs from all TSDL/SpecRL branches—learning cross-band and cross-scale interactions via lightweight channel-wise fusion, in order to formulate a unified, discriminative representation for the final softmax-based classification.

3.1. Channel-wise normalization

Prompted by the fact that EEG-based brain responses are often exposed to many possible external interferences, such as the subject's physiological activity, muscle movement (e.g. chewing, etc.), cardiac pulsations, or poor electrode contact etc, we observed and analyzed a large proportion of the randomized EEG-ImageNet [1], which indicates that there exist a noticeable proportion of artifacts, suspicious of abnormality for our proposed classification. Compared with the majority of EEGs, which demonstrates a relative consistency in their waveforms with statistics of signal intensities primarily concentrated in the mid-range of their max-

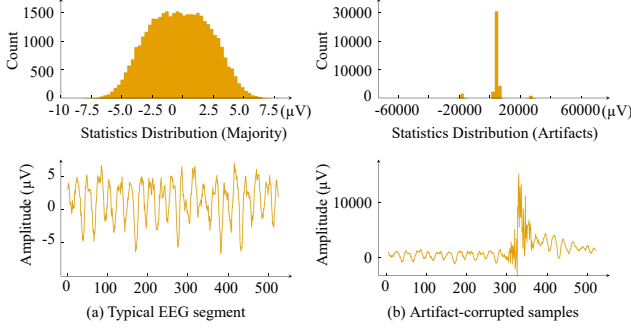


Figure 3. Comparative illustration of the observed EEGs

imum and minimum variations, these artifacts often illustrate prominent low and abrupt high magnitudes, exhibiting irregular fluctuations and deviating significantly from a normal distribution. Figure 3 illustrates a clear comparison between a typical EEG segment and an artifact-corrupted one. As shown in Figure 3(a), most EEGs in the dataset exhibit stable oscillatory waveforms with amplitudes generally within $\pm 6 \mu\text{V}$, and their statistics distribution can be characterized by a normal pattern. This reflects the regular rhythmic nature of neural activities. In contrast, Figure 3(b) demonstrates an artifact-contaminated EEG segment characterized by extreme amplitude excursions reaching tens of thousands of μV in an irregular manner with heavy-tailed value distribution.

To minimize such potentially negative impact upon our proposed TS-SpectrumNet, we introduce a channel-wise normalization to: (i) minimize the effect of possible artifacts as a result of that normalized EEGs fall closer to each other within similar numerical ranges across different channels; (ii) leverage the impact of individual differences and variations in signal strength; (iii) improve the coordination of individual feature learning across all channels.

Given the EEG sequence denoted as $\{x_{1c}, x_{2c}, \dots, x_{ic}, \dots\}$ $i \in [1, 512]$ and $c \in [1, 96]$ [1] or $[1, 128]$ [25], collected from channel c on the cap, our introduced channel-wise normalization, i.e. the normalized EEG sequence x'_{ic} , can be calculated via the following:

$$x'_{ic} = \frac{x_{ic} - \mu_c}{\sigma_c} \quad (1)$$

where μ_c and σ_c are the mean and the standard deviation of all EEGs for channel c , respectively.

3.2. TSDL: Temporal-Spatial Dynamics Learning

The temporal-spatial dynamics learning (TSDL) module is the first core processing stage of our network, applied directly after the channel-wise normalization of the raw 96×512 EEG input for randomized EEG-ImageNet and the raw 128×512 input for deterministic EEG-ImageNet. The primary objective of this module is to capture the rich

spatio-temporal dynamics inherent in EEG signals across multiple time scales simultaneously. As illustrated in Figure 2, the TSDL module employs a parallel, multi-branch architecture inspired by the "Inception" concept, allowing the network to learn from different temporal receptive fields concurrently [16].

The normalized input, $X_{norm} \in \mathbb{R}^{C \times T}$, is fed into three independent 1D convolutional branches, each with a different kernel size to sustain effective learning across the varying aspects of the distinct spatio-temporal dynamics, which can be highlighted as follows.

- **Small Kernel (S) Branch:** This branch utilizes 1D convolutions with a small kernel size ($k_S = 63$), in order to enable the learning module to capture high-frequency, local, and fine-grained spatio-temporal dynamics.
- **Medium Kernel (M) Branch:** This branch uses 1D convolutions with a medium kernel size ($k_M = 127$). This is aimed at enabling the dynamics learning to focus on those temporal dependencies and spatial correlations over medium sized time windows.
- **Large Kernel (L) Branch:** This branch is constructed by a number of 1D convolution units with a large kernel size ($k_L = 255$) to empower the dynamics learning with wider receptive fields, which is crucial for capturing low-frequency, long-range temporal dependencies and global spatio-temporal relationships across the EEG sequences.

As shown in Figure 2 (e.g., M1, M2, ..., M8), each branch consists of multiple parallel convolutional filters. The 1D convolution operation in each filter is "spatio-temporal" as it spans all C input channels while convolving along the temporal dimension T , thus learning patterns that involve correlations *between* channels *over* a specific duration. The outputs of the parallel filters within a single branch are then aggregated via element-wise summation (indicated by the symbol $\oplus$) to produce a single but robust feature map for that specific scale.

Specifically, for an input X_{norm} and a given branch $b \in \{S, M, L\}$ with N_b parallel filters ($N_b = 8$), the aggregated output feature map F_b is computed as:

$$F_b = \sigma \left(\sum_{i=1}^{N_b} \text{Conv1D}_{k_b}^{(i)}(X_{norm}) \right) \quad (2)$$

where $\text{Conv1D}_{k_b}^{(i)}$ is the i -th 1D convolutional filter with kernel size k_b , and σ represents the non-linear activation function (e.g., ReLU). The TSDL module finally outputs three distinct feature maps, F_S, F_M, F_L , which are then passed in parallel to three identical SpecRL modules for dedicated representational learning across different frequency domains.

3.3. SpecRL: Spectrum Representational Learning

It is known that CNNs have a limited receptive field, making it difficult to effectively capture global contextual in-

formation. This limitation is particularly problematic for classification of randomized EEG trials, where both local and global correlations need to be learned and exploited simultaneously, in order to ensure that every possible contribution is taken into full consideration. Correspondingly, we propose a new spectrum representational learning (SpecRL) layer, enabling efficient and effective capture of both local and global features to overcome the locality constraints across both the time and spatial domains of EEG trials.

As shown in Figure 2, the SpecRL layer firstly decomposes the input into different spectral components via two-level wavelet transforms, after which an inverse wavelet transform is applied to reconstruct the output and sustain a compatibility for information flow across all stages of the proposed framework. This mechanism is essentially designed to extend the receptive field during multi-resolution and multi-level decompositions, enhancing the model’s ability to learn spatial dynamics from EEG sequences while maintaining a low computational complexity. To decompose the input feature map into individual components across their spectrum, we adopt the Symlets wavelet kernel function to process the signal for better denoising and smoothing.

Specifically, given the input feature map, its j -th elements $X_L^{(i)}[j]$ and $X_H^{(i)}[j]$ can be decomposed via the following equations:

$$X_L^{(i)}[j] = \sum_{k=0}^{L-1} h_k \cdot X_L^{(i-1)}[2j+k] \quad (3)$$

$$X_H^{(i)}[j] = \sum_{k=0}^{L-1} g_k \cdot X_H^{(i-1)}[2j+k] \quad (4)$$

where, $X_L^{(i)}[j]$ and $X_H^{(i)}[j]$ represent the j -th reconstructed elements of the low-frequency and high-frequency components at the i -th level ($i \in [1, 2]$), respectively, and h_k and g_k are the coefficients of the low-pass and high-pass filters. L represents the filter length and $L_{Symlets}=4$. $X^{(i-1)}[2j+k]$ denotes the four consecutive samples starting from the $2j$ -th position in the input.

Upon each decomposed spectrum component, the introduced SpecRL layer performs small convolution operations, by which it is meant that smaller convolution kernels are compared with those traditionally large kernels, in order to help reducing the number of parameters and computational complexity. Essentially, this is designed not only to apply convolution to individual spectrum component, but also to allow small kernels to operate over a larger area of the original input, thereby maximizing the receptive field.

At each decomposition level i , convolution operations are applied to the decomposed low-frequency and high-frequency components individually, in order to maximize the role of the learned spectrum representation.

After completing the convolution operations, SpecRL uses the inverse wavelet transform (IWT) to combine the low and high-frequency components back to the original spatial resolution. This process preserves the integrity of both local and global information, fusing all contributions across the spectrum to form the final output.

Correspondingly, the n -th element of the reconstructed signal is obtained using the inverse wavelet transform, which is described below:

$$Y^{(i)}[n] = \sum_{k=0}^{L-1} h'_k \cdot \left(C_L^{(i)}[n-k] + Y^{(i+1)}[n-k] \right) + \sum_{k=0}^{L-1} g'_k \cdot C_H^{(i)}[n-k] \quad (5)$$

where, $Y^{(i)}[n]$ is the reconstructed signal at level i . $C_L^{(i)}$ and $C_H^{(i)}$ are the upsampled low-frequency and high-frequency components after convolution at level i . $Y^{(i+1)}$ is the aggregated output from the previous level $i+1$.

Finally, to coordinate the proposed learning modules and thus provide the best possible preparation for the classification layer, we further introduce a feature integration module (FIM) to optimally exploit the exposure of classifiable information and hence maximize the final classification effectiveness. As illustrated in Figure 2, the FIM takes the feature maps from all spectral branches as input. It then performs a comprehensive integration, where each output feature map is generated based on a learned combination of all input feature maps. This mechanism, often implemented as a channel-wise attention or a fully-connected layer across the feature dimension, allows the model to weigh the importance of different spectral components and hence capture their synergistic interactions.

4. Experiments and analysis

4.1. Experimental setup

To evaluate the proposed TS-SpectrumNet, we set up the experimental platform to follow two principles, i.e. the compatibility and fairness in comparisons; sufficiency and comprehensiveness in coverage. Details of such set up can be highlighted as: (i) for datasets, we primarily adopt the EEG-ImageNet with randomized EEG trials extended by Purdue researchers [1], and the EEG-ImageNet with deterministic EEG trials originated by Florida researchers [25]; (ii) for benchmarking, we not only include all the classifiers adopted for comparative experiments over randomized EEG trials [1], but also those representing the latest development in relevant areas, such as the transformer-based EEGConformer [23] and NICE-EEG, developed for brain decoding out of stimuli-EEG pairings [24].

Model	Randomized EEG Trials			Deterministic EEG trials		
	Top-1	Top-5	Params	Top-1	Top-5	Params
SVM[17]	5.0%	21.3 ± 0.9 (%)	1.9M	5.8 ± 0.3 (%)	18.2 ± 0.5 (%)	2.3M
KNN[17]	2.1%	12.1 ± 0.3 (%)	—	7.5 ± 0.3 (%)	26.9 ± 0.7 (%)	—
LSTM[7]	2.2%	12.4 ± 0.4 (%)	0.1M	21.8 ± 0.5 (%)	70.4 ± 2.1 (%)	0.1M
MLP[17]	2.5%	12.6 ± 0.4 (%)	6.3M	6.6 ± 0.2 (%)	25.6 ± 0.4 (%)	7.2M
CNN[17]	5.1%	21.4 ± 0.8 (%)	0.004M	32.1 ± 1.2 (%)	77.9 ± 1.8 (%)	0.006M
EEGNet[15]	7.0%	24.6 ± 1.1 (%)	0.01M	31.8 ± 1.0 (%)	90.5 ± 2.3 (%)	0.01M
[20]	2.5%	10.5 ± 0.3 (%)	4.5M	48.1 ± 1.4 (%)	82.9 ± 1.8 (%)	5M
SyncNet[18]	2.5%	13.8 ± 0.5 (%)	0.04M	33.8 ± 1.3 (%)	85.2 ± 2.0 (%)	0.06M
EEGConformer[23]	4.2 ± 0.2 (%)	17.5 ± 0.7 (%)	0.6M	45.3 ± 1.5 (%)	87.8 ± 1.8 (%)	0.6M
NICE-EEG[24]	5.1 ± 0.2 (%)	18.9 ± 1.0 (%)	0.3M	34.1 ± 1.0 (%)	78.1 ± 1.7 (%)	0.3M
EEG-TSSnet	15.1 ± 0.3 (%)	39.7 ± 1.3 (%)	0.05M	50.3 ± 1.6 (%)	91.9 ± 2.1 (%)	0.1M

Table 1. Comparative listing of experimental results.

Specific implementation of the proposed is completed in PyTorch and its execution is on an NVIDIA A100 GPU with five fold cross validation for both training and testing. The initial learning rate is set to 0.001, and the training epoch is 100. Other items include utilization of the Adam [14] optimizer and a cross entropy loss function.

4.2. Experimental results

Following extensive experiments carried out, we list all the classification results in Table 1, including Top-1 and Top-5 accuracy rates and the number of parameters required. While Top-1 and Top-5 results are adopted to measure the effectiveness of all the tested classifiers, the number of parameters are listed to measure their efficiency.

As seen, over the EEG-ImageNet with randomized trials [1], our proposed TS-SpectrumNet achieves a new state-of-the-art Top-1 accuracy at 15.1%, more than doubling the 7.0% from the previous best, EEGNet [1]. Our Top-5 accuracy of 39.7% also significantly leads the second-best EEGNet (24.6%), as well as other baselines, including transformer-based models, such as EEGConformer at 4.2%.

To maximize our coverage of comparative evaluations, we have also run all the experiments over the EEG-ImageNet with deterministic trials [25], the results of which are listed under the third column of Table 1. As seen, our proposed TS-SpectrumNet also achieves the best rate at 50.3%, outperforming the existing best EEGChannelNet [25] at 48.1%. In contrast, the best performing EEGNet on randomized EEG trials, however, achieved a classification rate of 31.8%, behind EEGConformer, NICE-EEG, SyncNet, and CNN.

To evaluate the efficiency of our proposed TS-SpectrumNet, we also list the results of parameter count for all tested models in Table 1, from which it can be seen that our proposed has delivered the best cost-effectiveness among all the tested models. We argue our case via the following observations: (i) For classification of deterministic EEG trials, proposed model delivers the best classification rate at 50.3% using only 0.1 million parameters. In contrast,

the second-best model, EEGChannelNet, achieves a rate of 48.1% with 5.0 million parameters. With less than 2% of the parameters, our proposed model achieves a 2.2% higher classification rate. (ii) For classification of randomized EEG trials, our proposed achieves the best rate at 15.1% upon 0.05 million parameters, yet EEGChannelNet only achieved 2.5% upon 4.5 million parameters, 90 times more than that of our proposed. (iii) Compared with EEGNet, the existing best performing model for classification of randomized EEG trials [1], the number of parameters required is 0.01 million, only 20% of our proposed. Considering the fact that our classification rate is more than doubling the rate by EEGNet, i.e. 15.1% against 7.0%, we argue that the price paid is justifiable. (iv) While NICE-EEG is well performed in the task of classifying images from image-EEG pairings [24] over the Gifford dataset [6], it is under performing for the task of classifying randomized EEG trials with a rate at 5.1%. For the task of classifying deterministic EEG trials, its achieved classification rate is 34.1%, under performing EEGChannelNet, TS-SpectrumNet, and EEGConformer.

In summary, from all the above evaluations over both the data sets with randomized and deterministic trials, it can be seen that our proposed TS-SpectrumNet delivers the best effectiveness and efficiency among all the compared baselines in terms of the overall considerations, including the magnitude of differences between the best rate and the second best rate, and combinational measurement of classification rate v.s. parameter count etc.

4.3. Ablation experiments

4.3.1. Module-wise ablation

To quantify the contribution of each architectural component, we conduct controlled ablation studies by removing one module at a time from TS-SpectrumNet while keeping all training protocols unchanged, and report Top-1/Top-5 accuracy rates on classifying both the randomized and deterministic EEG trials, details of which are summarized in Table 2. The four modules under the ablation studies are: (a) channel-wise normalization (Norm), (b) tempo-

Variant	Randomized EEG Trials		Deterministic EEG trials	
	Top-1	Top-5	Top-1	Top-5
w/o Channel-wise Norm.	14.7%	38.8%	49.1%	91.0%
w/o TSDL	12.8%	33.1%	45.4%	88.1%
w/o SpecRL	14.6%	38.5%	48.9%	90.8%
w/o FIM	14.8%	39.0%	49.6%	91.4%
TS-SpectrumNet (full)	15.1%	39.7%	50.3%	91.9%

Table 2. Module-wise ablation on EEG-ImageNet.

ral-spatial dynamics learning (TSDL), (c) spectrum representational learning (SpecRL), and (d) feature integration module (FIM).

From the results listed in Table 2, the following four observations can be made: (i) removal of channel-wise normalization incurs degradation across all classification rates, indicating that introduction of this scheme not only improves the stability of performances for our proposed deep framework, but also maintains a better inter-channel balance and scale alignment towards the quality improvement of input EEG trials; (ii) removal of TSDL incurs the largest degradation (15.1%→12.8% Top-1 on randomized trials, and 50.3%→45.4% Top-5 on deterministic trials), validating that TSDL is playing a pivotal role in classification of EEG trials, and temporal-spatial dynamics represents the largest proportion in characterizing EEG trials; (iii) removal of SpecRL consistently reduces all classification rates across both randomized and deterministic trials, confirming its effectiveness in representational learning across Wavelets decomposition of multi-resolution spectrum beyond local constraints; (iv) although removal of FIM incurs the smallest degradation in classification rates, its role in coordinating all different modules inside our proposed deep framework is still positive and important, validating that learning from cross-branch interactions helps sustaining the discriminative features and hence finalizing the overall classification performances.

4.3.2. Effect of Channel-wise Normalization

To validate our channel-wise normalization (Section 3.1), we illustrate the experimental results in Top-1 rates achieved by all compared models in Figure 4, where Raw represents unnormalized. As seen, the majority of compared models (8 out of 11) achieved better rates with normalization. Notably, EEGNet improved from 7.0% to 7.4%, while our proposed TS-SpectrumNet saw the largest gain, jumping from 14.7% to 15.1%. These results confirm that channel-wise normalization is a crucial and beneficial pre-processing step, effectively aiding classifiers in focusing on relevant EEG patterns by reducing noise interference.

4.3.3. Effect of convolution kernel size

To assess the effect of multi-scale convolutions for the temporal-spatial dynamics learning module, we carried out

Scheme	Kernel 1	Kernel 2	Kernel 3	Accuracy
Scheme 1	5×5	3×3	1×1	12.1%
Scheme 2	64×64	32×32	16×16	12.9%
Scheme 3	1×64	1×32	1×16	14.5%
Scheme 4	1×255	1×127	1×63	15.1%

Table 3. Test results on effect of kernel sizes.

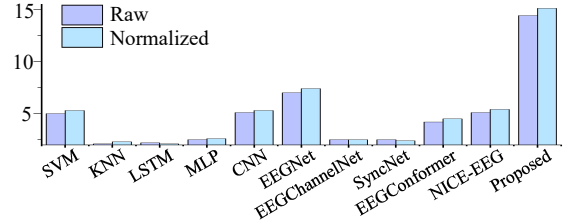


Figure 4. Results of removing the effects of artifacts on classification.

ablation experiments with varying combination of kernel sizes and list their results in Table 3. From these results, the following observations can be made: (i) Combination of smaller kernel sizes underperforms as their receptive field is limited to capturing short-term dependencies, leading to insufficient capture of long-term temporal dynamics across EEG sequences along the time dimension. (ii) Larger kernels, such as 64×64 , 32×32 , 16×16 , tend to cover a broader area and hence are capable of capturing more temporal information. However, due to the fact that we need to simultaneously learn both spatial and temporal dynamics and their interaction remains crucial, a lack of strong focus on temporal dynamics could create a negative impact upon their performances. (iii) The combination of 1×64 , 1×32 , 1×16 delivered significant performance improvement, validating that their structural emphasis upon the temporal dimension played positive roles. (iv) Further increasing the kernel sizes along the temporal dimension, as shown by the final row in Table 3, rendered additional improvement, delivering the best performance. Other combinations beyond this optimal point, however, started to produce negative impacts upon their performances.

4.3.4. Effect of wavelets

To assess the effect of multi-resolution decomposition for the proposed spectrum representational learning module, we carried out ablation experiments over a range of widely adopted wavelets, including Reverse Biorthogonal, Biorthogonal, Coiflet, Haar and Symlet. The results, shown in Figure 5, reveal significant differences in classification accuracy among these wavelet kernels. Simpler wavelet kernels, such as Reverse Biorthogonal and Biorthogonal, perform poorly in handling complex signals, while the Haar wavelet, despite its computational efficiency, is limited in

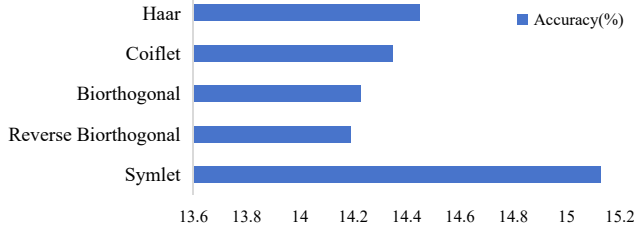


Figure 5. Test results on the effect of Wavelets.

processing smooth signals due to its step-like structure, making it challenging to capture subtle variations in the signal.

In contrast, as shown in Figure 5, the Symlet wavelet kernel performed the best, achieving a classification accuracy of 15.1%, which is notably higher than the other wavelets. The advantage of symet lies in its symmetry, which not only enhances robustness against artifacts but also improves noise filtering, allowing it to capture key representation more precisely in complex signal environments. Additionally, the time-frequency localization property of Symlet enables it to effectively capture both high and low-frequency components, providing better resolution for non-stationary signals. Overall, the smoothness and spectral characteristics of Symlet make it particularly effective in EEG signal processing, underscoring the importance of selecting an appropriate wavelet kernel to enhance classification accuracy.

4.3.5. Effect of label cardinality

To assess the scalability of our proposed model with respect to the task granularity, we systematically vary the number of target classes K and examine how the Top-1 classification rates evolve changes as space of labels expands, and the results are summarized in Table 4. As K increases from 1 to 40, both the compared models, EEGNet and TS-SpectrumNet, exhibit a smooth drop in Top-1 accuracy, indicating a well-behaved evaluation protocol that reflects the expected rise in difficulty. Across all K , TS-SpectrumNet consistently outperforms EEGNet, and the margin widens with larger K . For example, at $K=2$, the accuracies are 75.5% vs. 50.0%, at $K=10$ they are 33.5% vs. 17.3%, and at $K=40$ they are 15.1% vs. 7.0%. Measured against the chance rate $1/K=40$, i.e. 2.5%, our method reaches about 6 times the chance rate, whereas EEGNet reaches about 2.8 times the chance rate, suggesting that the proposed TS-SpectrumNet preserves stronger inter-class separability and generalizes more stably as the number of classes grows.

5. Conclusion

In this paper, we have reported our latest research on classifying brain perception of natural images from both the randomized and deterministic EEG trials with a new deep

#Classes	Method		#Classes	Method	
	EEGNet	Ours		EEGNet	Ours
1	-	-	21	12.0+0.8	21.1+0.8
2	50.0+1.3	75.5+3.4	22	11.7+0.1	20.4+0.0
3	35.1+0.5	62.3+0.0	23	11.2+0.2	19.7+0.1
4	30.2+1.0	51.8+0.9	24	10.9+0.3	19.2+0.4
5	24.8+1.0	45.2-1.2	25	10.7+0.5	18.7+0.0
6	21.8+1.0	41.8+0.2	26	10.5+0.4	18.3+0.2
7	19.5+0.3	37.6+0.6	27	10.1+1.0	17.9+0.4
8	17.4+0.3	35.3+0.7	28	9.9+0.7	17.3+1.2
9	18.2+0.1	34.4-0.1	29	9.7+0.5	17.0+0.7
10	17.3-0.2	33.5+0.0	30	9.9+0.8	16.6+0.5
11	17.3+0.3	34.2-0.1	31	9.3+0.6	16.2+0.4
12	16.6+0.7	31.2+0.7	32	9.1+0.5	15.9+0.1
13	15.6+0.6	30.4+0.3	33	8.8+0.5	15.8+1.2
14	15.2+1.0	29.7+0.3	34	8.6+0.4	15.9+0.5
15	15.7-0.4	28.8-0.2	35	8.6+0.8	15.6+0.9
16	15.4+0.4	27.3-0.2	36	8.6+0.5	15.6+0.4
17	14.4-0.4	26.0+0.5	37	8.0+0.8	15.5+0.4
18	13.9+0.5	24.7-0.6	38	7.6+0.4	15.3+0.4
19	13.4+1.0	23.0+0.5	39	7.4+0.4	15.1+0.5
20	12.7+0.3	21.6-0.1	40	7.0+0.6	15.1+0.7

Table 4. Test results on the effect of artifact removal across the number of classes.

framework, TS-SpectrumNet. From our comparative investigations and experimental analysis, it can be concluded that: (i) all the tested models suffer from significant drops in classification rates from deterministic EEG trials to randomized EEG trials, and the maximum classification rate is now pushed forward by our proposed TS-SpectrumNet from the existing maximum rate at 7.0% to the reported rate at 15.1%, indicating that classification of brain responses from randomized EEG trials is indeed challenging. (ii) TS-SpectrumNet achieves state-of-the-art performances on both randomized and deterministic EEG trials, delivering the best Top-1 and Top-5 classification rates in comparison with all the representative existing baselines. (iii) the heterogeneous behavior across baselines suggests that a single inductive bias may not serve both settings equally well. For example, EEGChannelNet attains only 2.5% on randomized trials but remains the best among all existing models on deterministic trials (48.1%). These observations highlight the importance of explicitly modeling temporal-spectral structure and interactions, which our proposed leverages to generalize robustly across evaluation protocols.

Further research can be identified in three directions, including: (i) promoting the quality or increasing the level of classifiable information quantity for randomized EEG trials is proved to be important by our investigations in this paper; (ii) multi-modal learning, such as audio/visual stimuli or combination of speech/image etc., may provide additional new solutions; and (iii) knowledge transfer [31] or zero-shot learning [28] across EEG domain and stimuli domain may prove worthwhile for further investigation, as both domains share certain level and type of correlations.

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